

## EXECUTIVE SUMMARY

Week 6 builds on Week 5 by moving from descriptive analysis to advanced analysis, validation, segmentation and decision support. The overall no-show rate remains 48.46%, while attendance is 46.28%.

The strongest risk factors are previous no-shows and long booking lead time. Appointments booked 31+ days in advance have a 60.49% no-show rate, while the identified high-risk segment has a 61.02% no-show rate across 233 appointments.

## ADVANCED ANALYSIS

The combined analysis of previous no-shows and booking lead time showed that patients with repeated previous no-shows and long lead times are particularly high-risk, with some combinations reaching approximately 80% no-show rates.

Reminder analysis showed lower no-show rates among patients who received reminders across all previous no-show groups. For example, patients with 3+ previous no-shows recorded approximately 76.00% without reminders compared with 66.18% with reminders. However, this is an association and does not prove causation.

Long-lead appointments also remained high-risk across distance groups. Appointment day, appointment time and age showed smaller differences and appear less important than previous no-shows and booking lead time.

## Key KPIs

The Week 6 KPIs include:

- **Total Appointments:** 5,000
- **No-Show Rate:** 48.46%
- **Attendance Rate:** 46.28%
- **Cancellation Rate:** 5.26%
- **Reminder Coverage:** 72.68%
- **Long-Lead No-Show Rate:** 60.49%
- **High-Risk No-Show Rate:** 61.02%
- **High-Risk Appointments:** 233
- **Reminder No-Show Gap:** 4.03 percentage points

## **KEY INSIGHTS**

1. Previous no-shows are the strongest behavioural warning signal.
2. Long booking lead time is a major operational risk factor.
3. Combining risk factors provides stronger decision support.
4. Reminders are associated with lower no-show rates.
5. Previous no-shows, lead time and distance are stronger indicators than appointment day and time.

## **RECOMMENDATIONS**

HealthConnect should adopt a risk-based appointment support strategy. Patients with 3+ previous no-shows and appointments booked 31+ days in advance should receive additional attention and follow-up.

Reminder coverage should be increased for appropriate high-risk appointments, while the effectiveness of different reminder channels should be investigated.

HealthConnect should also consider developing a future risk score using previous no-shows, booking lead time and distance and investigate transportation or accessibility barriers for patients living farther from the clinic.

The Sunday appointment inconsistency should be resolved before Sunday findings are used for operational decisions.

## **DASHBOARD AND DATA SCIENCE CONTRIBUTION**

The Week 6 Power BI dashboard was enhanced with risk KPIs, Previous No-Shows × Booking Lead Time, reminder-effect analysis, long-lead × distance analysis and monthly validation.

The analysis provides potential predictive features for Data Science, including previous no-shows, booking lead time, reminder status, distance, appointment type and age. Interaction effects should also be considered in future modelling.

## **LIMITATIONS**

The analysis is affected by missing values and the unresolved Sunday data issue. The findings show associations rather than causation, and the high-risk segment should be used for prioritisation rather than as a definite prediction of individual behaviour.

## **WEEK 7 REQUIREMENTS**

Week 7 should statistically test the reminder effect, previous no-shows and lead time, investigate interaction effects, validate the high-risk segment, examine monthly stability, resolve the Sunday issue.

## **CONCLUSION**

Week 6 confirms that previous no-show behaviour and long booking lead time are the strongest risk indicators in the HealthConnect dataset. The 61.02% high-risk no-show rate demonstrates the potential value of targeted appointment support.

The key recommendation is to move toward risk-based appointment management, focusing on combinations of previous no-shows, booking lead time, reminders and accessibility factors. These findings provide a strong foundation for Week 7 statistical testing and predictive modelling.